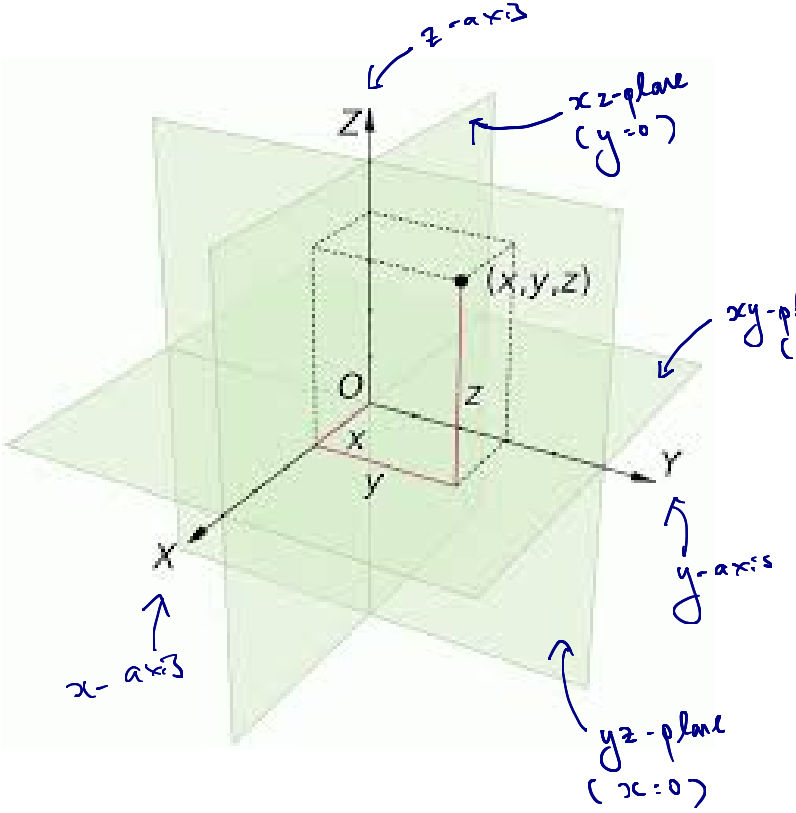
11.1 Three-Dimensional Coordinate Systems



"Space" = 3-dimensional space

A point is uniquely determined by its x-, y-, z-components

A point in space is usually written as $P = P(x, y, z)$.

Remark:

1. The Cartesian coordinates for space are also called **rectangular coordinates** because they have three perpendicular axes.
   a. The x-axis has the points with coordinates of the form $(x, 0, 0)$.
   b. The y-axis has the points with coordinates of the form $(0, y, 0)$.
   c. The z-axis has the points with coordinates of the form $(0, 0, z)$.

2. The coordinate axes define 3 **coordinate planes**:
   a. The xy-plane, whose standard equation is $z = 0$ and the set of all points in the plane is
   $$\{(x, y, 0) \mid x \in \mathbb{R},\ y \in \mathbb{R}\}.$$
   b. The yz-plane, whose standard equation is $x = 0$ and the set of all points in the plane is
   $$\{(0, y, z) \mid y \in \mathbb{R},\ z \in \mathbb{R}\}.$$
   c. The xz-plane, whose standard equation is $y = 0$ and the set of all points in the plane is
   $$\{(x, 0, z) \mid x \in \mathbb{R},\ z \in \mathbb{R}\}.$$

These three coordinate planes divide space into **8 octants**. The octant in which the point of coordinates are positive is the **first octant** and there is no conventional numbering for the other 7 octants. First octant : $\{(x, y, z) : x \geq 0, y \geq 0, z \geq 0\}$

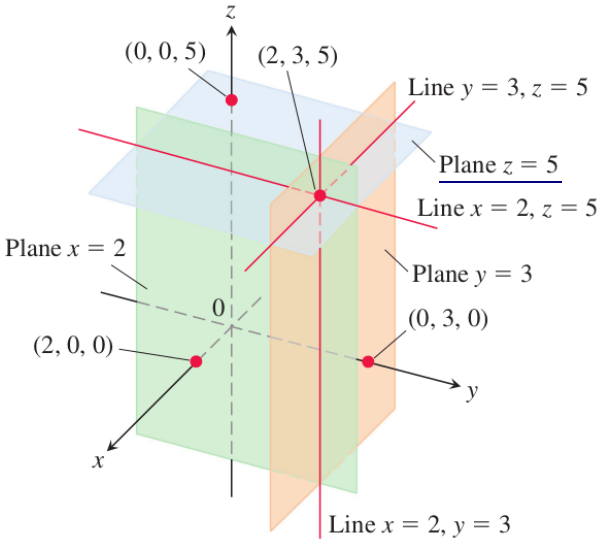


FIGURE 11.3 The planes $x = 2$, $y = 3$, and $z = 5$ determine three lines through the point (2, 3, 5).

3. Let $k \in \mathbb{R}$.

   a. The equation of the plane parallel to the xy-plane is $z = k$.

   b. The equation of the plane parallel to the yz-plane is $x = k$.

   c. The equation of the plane parallel to the xz-plane is $y = k$.

Example:

Describe the following sets of points in space.

1. $\{(0, y, 0) \mid y \in \mathbb{R}\}$

   The y-axis

2. $\{(0, y, z) \mid y \in \mathbb{R},\ z \in \mathbb{R}\}$

   The yz-plane.

3. $\{(x, y, -2) \mid x \in \mathbb{R},\ y \in \mathbb{R}\}$

   The plane parallel to xy-plane and 2 units below the xy-plane.
   (Can also be described as the plane with equation $z = -2$.)

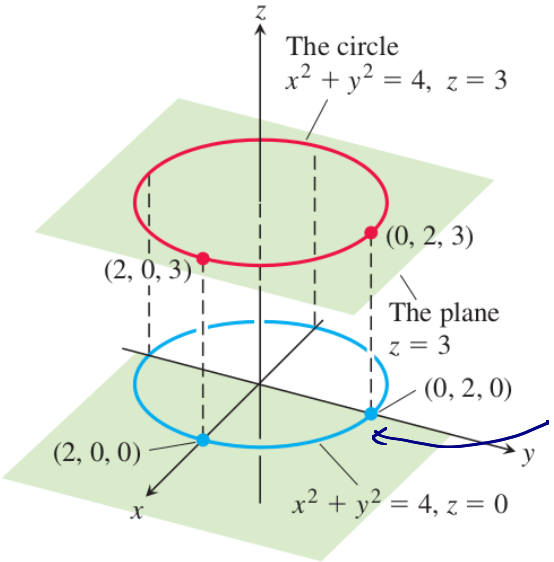
4. $\{(1, 2, z) \mid z \in \mathbb{R}\}$

   The intersection line of the planes $x = 1$ & $y = 2$.

Example: The points $P(x, y, z)$ satisfying the equations

$$x^2 + y^2 = 4 \qquad \text{and} \qquad z = 3$$

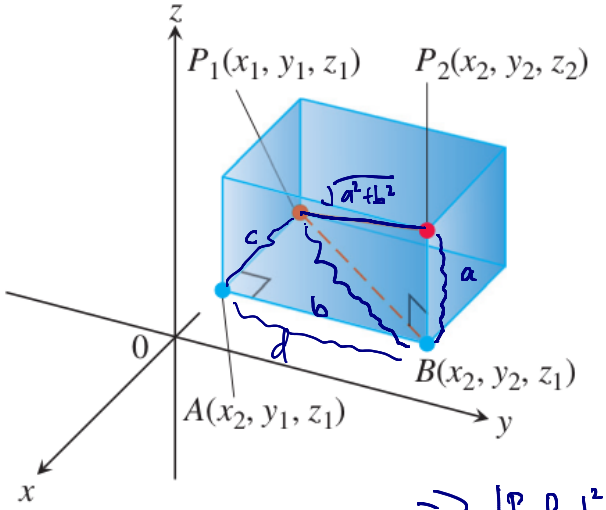
form a circle in the horizontal plane $z = 3$.



$x^2 + y^2 = 4,\ z = 0$

FIGURE 11.4 The circle $x^2 + y^2 = 4$ in the plane $z = 3$ (Example 2).

The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$



$|P_1P_2|^2 = |P_1B|^2 + |BP_2|^2$ ($\because$ $\triangle P_1BP_2$)

$|BP_2|^2 = (z_2 - z_1)^2$

$|P_1B|^2 = |AP_1|^2 + |AB|^2$

$= (x_2 - x_1)^2 + (y_2 - y_1)^2$

$\Rightarrow |P_1P_2|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$

$\Rightarrow |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

The collection of points with the same distance from a particular point forms a sphere.

The Standard Equation for the Sphere of Radius a and Center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

↑ All the points with distance a from the point (x_0, y_0, z_0).

Example: Find the center and radius of the sphere

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0.$$

$$(x^2+3x) + y^2 + (z^2-4z) + 1 = 0$$

$$(x+\tfrac{3}{2})^2 - \tfrac{9}{4} + y^2 + (z-2)^2 - 4 + 1 = 0$$

$$\Rightarrow (x+\tfrac{3}{2}) + y^2 + (z-2)^2 = \tfrac{21}{4}$$

∴ The centre is @ $(-\tfrac{3}{2}, 0, 2)$ and the radius is $\sqrt{\tfrac{21}{4}} = \tfrac{\sqrt{21}}{2}$.

Example:

Let S be the sphere of radius 2 centred at the origin.

↳ $x^2+y^2+z^2=4$

Interpret the following equations and inequalities.

1. $x^2 + y^2 + z^2 < 4$ - *the interior of the sphere S*

2. $x^2 + y^2 + z^2 \leq 4$ - *the interior of the sphere S, together with S*

3. $x^2 + y^2 + z^2 > 4$ - *the exterior of the sphere S*

4. $x^2 + y^2 + z^2 = 4,\ z \leq 0$ - *the lower hemisphere cut from the sphere S by the xy -plane*

